INF211

PROJECT-2

FOR STUDENTS

Project Description: 2D Simulation Game

Objective

The goal of the game is to move a character from the starting point to a designated target **(H)** on a square grid (4x4, 6x6 etc.), while avoiding obstacles **(X)**. Players must make correct decisions to successfully reach the target without colliding with obstacles.

Game Features

- A square grid is generated, serving as the game board and **has arbitrary size**. Starting point of the character is **(0,0)** and goal is reaching target (H) at **the last index of the board** without colliding obstacles.
- The grid dynamically updates with the player's movements, obstacles, and the target.

Interactive Gameplay

- Players control the character using commands as following:

(up → "u" , down → "d", left → "l", right → "r")

(upper right → "ur" or "ru", upper left → "ul" or "lu", down right → "dr" or "rd", down left → "dl" or "ld")

- If player reaches target, a win message must be printed and if player hits an obstacle, a game over message must be printed. Either way program must be terminated and game must be ended.

Obstacle Creation

- Obstacles must be placed randomly and total of their number is equal to the size of the board (i.e. number of obstacle = 4 for a 4*4 grid). You must ensure that these random created obstacles doesn't trigger a certain loss condition.

(Tip: random.randint() command from random library is advised to be used.)

Win/Loss Conditions

- Victory: Reach the target (H).
- Defeat: Collide with an obstacle (X).

Things you should be careful with:

- Ensuring that obstacle placement does not create a certain loss condition is one of the most important things in the project. You are expected to be creative and take advantage of the fact that we have a limited number of obstacles, that is, equal to one side of the grid. Therefore you can set certain constraints to obstacles that eliminate the possibility of a certain loss as given below in the figures. Aforementioned conditions must be applicable to any board size. **If you choose to use existing pathfinding algorithms to solve this problem rather than creating something new, such as BFS (Breadth-First Search) or DFS (Depth-First Search) algorithms, you are expected to explain the used algorithm very well. In other words, you must know not only what it does but also how it works.**

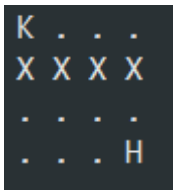
- You are forbidden to use built-in functions other than mentioned ones. For example, **you can only use random.randint() or a similar function for random obstacle generating.**

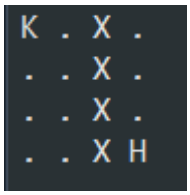
- The project is mainly about lists, so you are advised to use list methods such as **.join()** , **.split()** etc.

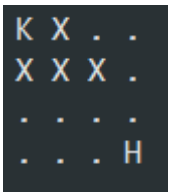
- Be careful with the borders. In other words, the character cannot go out of the board. When a obstacle is hit, the player must be erased from the board. You don't need to erase the player when reached target.

- Code must work in a single run

- The board must be created by lists.







How to play:

Losing Scenario

```
2D Simülasyon Oyununa Hoşgeldiniz!
Oyunun başında bir karakter seçeceksiniz.
Hedefe (H) ulaşmaya çalışın, engele (X) çarpmamaya dikkat edin.
Komutlar: hareket için: [ u / d / l / r etc.], çık
Lütfen bir karakter seçin (ör: K): K
K . . X
. . . .
. X . X
. . X H

Komut: d
. . . X
K . . .
. X . X
. . X H

Komut: dr
Engele çarptınız! Oyunu kaybettiniz.
. . . X
. . . .
. X . X
. . X H

Oyun bitti. Teşekkürler!
```

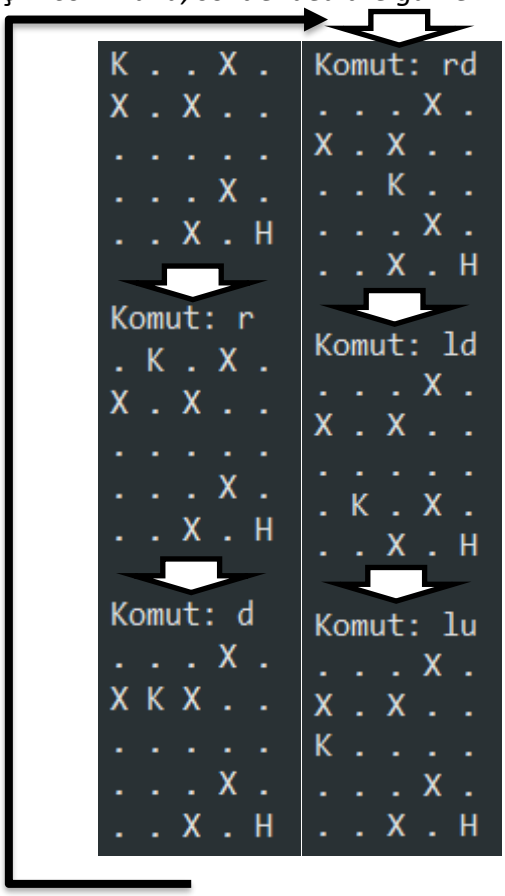
Explanation: First, we moved down then down right. Since there is an obstacle there, our character got removed from the game and the game has ended.

Winning Scenario

```
2D Simülasyon Oyununa Hoşgeldiniz!
Oyunun başında bir karakter seçeceksiniz.
Hedefe (H) ulaşmaya çalışın, engele (X) çarpmamaya dikkat edin.
Komutlar: hareket için: [ u / d / l / r etc.], çık
Lütfen bir karakter seçin (ör: K): K
K X . X
. . . .
. X . .
X . . H

Komut: dr
. X . X
. K . .
. X . .
X . . H

Komut: dr
. X . X
. . . .
. X K .
X . . H

Komut: dr
Hedefe ulaştınız! Tebrikler, oyunu kazandınız!
. X . X
. . . .
. X . .
X . . H

Oyun bitti. Teşekkürler!
```

Explanation: We reached the H spot without hitting any obstacles, so we got congratulations message. Also, program has stopped working with a message of game over!

Quitting Scenario and Movement Example

```
2D Simülasyon Oyununa Hoşgeldiniz!
Oyunun başında bir karakter seçeceksiniz.
Hedefe (H) ulaşmaya çalışın, engele (X) çarpmamaya dikkat edin.
Komutlar: hareket için: [ u / d / l / r etc.], çık
Lütfen bir karakter seçin (ör: K): K
K . X .
X . . X
. . . .
. . X H

Komut: çık
Oyun kapatılıyor. Teşekkürler!
```

We entered the 'çık' command, so it ended the game without losing or winning.